No of Pages :4 **Course Code : 23MX12**

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, DECEMBER 2024

MCA Semester: 1

23MX12 STRUCTURED PROGRAMMING CONCEPTS

Time : 3 Hours **Maximum Marks :100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Course Outcome : Table

Qn.1	CO1.	Qn.2	CO2.	Qn.3	CO3.	Qn.4	CO4

COs	At the end of learning this course, students will be able to
CO1	Choose appropriate Programming Paradigm, language and environments for problem solving and program development.
CO2	Analyse language constructs of imperative paradigm for program development.
CO3	Organize the software applications by analyzing and using proper structured solutions in Coding and data structures.
CO4	Develop programs using programming constructs to solve problems with emphasis on best practices like SCM, Patterns and Frameworks opted

		Question		(BTL)
1.	a)	Compare and contrast top down approach and bottom approach for structured programming.	(3)	(L4)
	b) i)	Demonstrate with suitable illustration passing arguments at runtime?	(5)	(L3)
	ii)	Analyze the code given below and write atleast 5 test cases.	(5)	

```
#include <stdio.h>
#include <stdlib.h>
int main(intargc, char *argv[]) {
    if (argc != 3) {
        printf("Insufficient Arguments!");
        return 1;
    }
    int num1 = atoi(argv[1]);
    int num2 = atoi(argv[2]);
    int sum = num1 + num2;
    float quotient = (num2 != 0) ? (float)num1 / num2 : 0.0f;
    printf("Sum: %d\n", sum);
    printf("Quotient: %.2f\n", quotient);
```

```
    return 0;
}
```

c) Discuss various characteristic features for classification of programming languages. Identify the position of C programming language in classification chart. For what type of applications C language shall be used for software development? (12) (L3)

2. a) Demonstrate a situation to use a while statement where a for loop would not be appropriate. (3) (L3)

Convert the following for-loop into an equivalent while-loop

```
for(i = 2; i<= sqrt(n); i += 3)
printf("%d ", i);
```

b) i) Analyze the scope and lifetime of the following variables, providing examples for each (a) External variable (b) Static variable (c) Automatic variable (d) Register variable. (6) (L4)

ii) Trace and Write the output for the given code. (4)

```
#include <stdio.h>
  void call();
  int main()
  {
    call();
    call();
  }
  void call()
  {
    inti = 11;
    int k=0;
    printf("%d ", i);
    static int j = 12;
    j = j + 1;
    k = ++j;
    printf("%d%d\n", j,k);
  }
```

c) i) Discuss all decision making and iterative statements with their syntax, semantics and sequence controls in C programming language with suitable illustrations. (8+4)

Trace and Interpret the output of the following program. (L4)

```
#include <stdio.h>
int main()
{
inti = 1, j = 1, k = 1, count = 0;
while (i< 2) {
for(; j < 4; j += k)
```

```
do {
++count;
k += i;
} while (k < 8);
i += j;
}
printf("Loop Indices: %d %d %d\n", i, j, k);
printf("Number of iterations = %d\n", count);
return 0;
}
```

(OR)

ii) Discuss the various components and types of user defined function? (L4)

What could happen if a recursive function lacks a proper base case or if the base case is never reached? Give a suitable example. Analyze the code step-by-step and write the output. (6+6)

```
#include <stdio.h>
void trickyRecursion(int *arr, int n) {
   if (n == 0) {
      return;
   }
   *arr += n;
   trickyRecursion(arr + 1, n - 1);
   *arr += n;
}
int main() {
   intarr[] = {1, 2, 3, 4, 5};
   int size = 5;
   trickyRecursion(arr, size);
   for (inti = 0; i< size; i++) {
      printf("%d ", arr[i]);
   }
   return 0;
}
```

3. a) What do you mean by enumeration in C and how does it help in code readability and maintability. Trace and find the output of the code snippet below. (L2) (3)

```
#include <stdio.h>
enumdesignFlags{ CROP = 1,ROTATE = 2,SAVE = 4};
int main() {
int myExample = ROTATE | SAVE | CROP;
printf("%d", myExample);
```

```
return 0;
}
```

b) Analyze the different ways of initializing string? Explain any five string functions used in C? (6+4) (L4)

Write a C program to implement the following You are given two strings s1 and s2 of equal length. A string swap is an operation where you choose two indices in a string (not necessarily different) and swap the characters at these indices. Return true if it is possible to make string S_1 same as string S_2 by performing at most one string swap Otherwise, return false. (L6)

c) i) 1) Differentiate static and dynamic memory allocation and Discuss the functions used in dynamic memory allocation. (6) (L2)

2) Write a C program that dynamically allocates memory for a 2D array of strings, takes input from the user, and uses pointer arithmetic to identify and display the minimum element in the array. (6) (L6)

(OR)

ii) Explain the difference between the 'r' and 'r+' modes when opening a file in C. Provide an example for each mode to demonstrate their use. (L2)

Write C program that accomplishes the following. Consider the structure (6+6)

```
Struct item{
  int id;
  char name[20];
  int quantity;
  int cost;          };
```

Assume that the file “iteminfo” already has 5 records. (L6)

a. Write a program that opens the file and appends the given new record to the file.

b. Display the content of the file.

4. a) Analyze and identify the situation when to use scripting language and system programming for application development. (3) (L4)

b) i) Discuss the features of markup languages in web Programming. (5) (L3)

ii) Explain the concept of static and dynamic libraries. (5)

c) Compare multi-paradigm languages with domain-specific languages, and also discuss when would you choose one over the other with typical use cases. (12) (L4)

CO : COURSE OUTCOME; BTL : BLOOM’S TAXONOMY LEVEL

/END/

FD/RSP